# Pedestrian and Vehicle Area Positioning With Multiple UWB Signals

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***Abstract*—In industrial environments where pedestrians and vehicles closely interact, existing point positioning systems often yield inaccurate results by disregarding crucial vehicle information, such as shape and angles, thereby compromising effective early warning mechanisms. Moreover, existing positioning systems relying on visual or radar methodologies also have limitations due to environmental factors. To address these challenges, this article introduces a novel pedestrian and vehicle area positioning and warning system base on multiple ultrawideband (UWB) signals, offering detailed information about vehicle positioning, orientation, and shape. Employing multiple UWB tags alongside the two-way ranging (TWR) algorithm, this system models the vehicle's positioning area and triggers alerts when the pedestrian-to-vehicle distance falls below a specified threshold derived from the shape and orientation data of the positioning area. Additionally, this work proposes a priori data for coordinate calibration, integrating extended Kalman filtering (EKF) and a dynamic threshold algorithm to seamlessly recalibrate the vehicle's positioning in both stationary and mobile scenarios. Experimental results demonstrate that, compared with using original area data for positioning, the modified algorithm can reduce the positioning coordinate STD by 18.43%, and the vehicle body shake variance by 50.56%, while also achieving a 4.30% increase in success warning rate and a notable 33.35% decline in false warning rate.**

***Index Terms*—IoT, Kalman filtering, positioning system, two-way ranging (TWR), ultrawideband (UWB).**

## I. Introduction

THE DEMAND for spatial positioning is constantly increasing, especially in the industrial sector [1]. In complex scenarios like chemical plants, workshops, and those interact frequently with vehicles, such as forklifts and cars, precise positioning of pedestrians, and vehicles, helps to monitor and detect abnormal situations in real time [2]. Utilizing positioning information for warning can be issued to prevent potential dangers.

As technology advances, spatial positioning methods like ultrawideband (UWB), Zigbee, RFID, and Wi-Fi are gaining prominence [3], [4], [5], [6], [7]. In the complex scenarios mentioned above, positioning based on UWB has advantages over other indoor positioning system (IPS) due to its centimeter-level accuracy, high-temporal resolution, low-power consumption, and excellent penetration ability [8]. Currently, the primary focus of research in UWB positioning is directed toward improving the robustness and precision of the positioning methodology. For example, researchers have focused on improving accuracy by utilizing the spacing of multiple positioning tags as a constraint condition, filtering redundant distance settings' weight values in [9], and the gradient descent method is used to optimize the initial coordinate value calculated by the Cafferty method, enhancing the positioning accuracy. Concurrently, efforts are made to bolster the accuracy and stability of positioning systems. In [10], a dynamic reference anchor Time Difference of Arrival (TDoA) compensation algorithm is proposed to enhance accuracy beyond anchor coverage by eliminating local minima of the TDoA loss function near anchors. Guido Schroeer et al. proposes a TDoA positioning system in [11] which employs four transceivers for all anchors instead of just one, thus improving the system robustness. The evaluation of multipath and non Line-of-Sight (NLoS) situations is conducted based on channel effects and their impact on positioning accuracy. A machine learning-based calibration method proposed in [12] aims to reduce positioning errors arising from reference inaccuracies in anchor positions. The accurate UWB positioning system, VULoc, outlined in [13], demonstrates high scalability for a variable number of targets and significantly reduces synchronization and calibration costs. Moreover, the maximum

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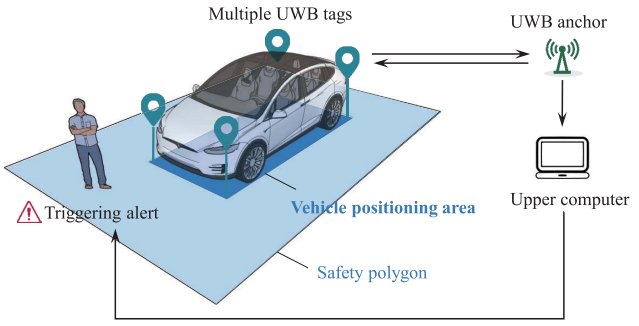



Fig. 1. Pedestrian and vehicle area positioning and warning system structure.

detection distance of tags on forklifts in complex scenarios is discussed in [14], proving the feasibility of applying pedestrian and vehicle positioning systems in complex scenes through the rational deployment of anchors and onboard tags.

Furthermore, multitag positioning technology has been commonly employed in positioning systems. In [15], a hexacopter is equipped with three UWB tags in a strategic way, and in [16], the utilization of multiple UWB tags on a mobile platform enables an agent to approach and land through the integration of range and vision measurements, but the accuracy of the positioning is constrained due to the absence of orientation information. A scenario of multitag area positioning is proposed in [17], where three mobile tags are fixed on the same handcart. The moving handcart records the positions of each tag at every moment, leading to approximately 80% availability performance. However, due to the neglect of overall information about the positioning area and optimization of trajectory and steering, the availability remains relatively low. In the study presented in [18], a framework for relative positioning and orientation estimation based on graph optimization and UWB distances is introduced. However, the positioning and orientation estimation rely on information provided by tags attached to other positioning targets, offering only relative positioning.

Presently, industrial UWB positioning systems primarily focus on accurately determining the coordinates of individual points. In [8], researchers have evaluated the performance of UWB-based industrial positioning systems deployed in real-world industrial scenarios, considering both static and mobile usage. However, the situations mentioned in this article are limited to determining the positioning of each asset using a single tag. Besides, various scenarios within current UWB industrial applications are discussed in [19]. These scenarios include mobile tag positioning, collaborative positioning among multiple robots or drones, and adding location metadata such as UWB gateways. However, the predominant focus in these applications is on point positioning and ranging. Consequently, there is a deficiency in absolute information concerning the shapes and angles of regions within the scenes, and real-time warning applications leveraging UWB positioning information have yet to be explored.

To address these constraints, we propose a multiple UWB positioning and warning system for precise positioning of pedestrian and vehicle area. Illustrated in Fig. 1, vehicle positioning area refers to locating the polygonal area where the vehicle is located. The proposed system focuses on solving the overall positioning and optimization of vehicle area, and shows superiority in: 1) the positioning is achieved solely using UWB multiple tags, allowing for complete decoupling from other communication devices; 2) it exhibits excellent performance for both stationary and moving vehicles, realizing a robust UWB warning application; and 3) providing vehicle positioning, orientation, and shape information, along with corresponding optimized calibration algorithms, has facilitated the achievement of high-performance positioning and warning services. In addition, this system has excellent application prospects, which can be applied to precise automatic parking, automatic unlocking, etc.

The main contributions of this article are as follows.

1) We propose a novel multitag UWB pedestrian and vehicle area positioning and warning system, which creatively introduces the concept of area positioning that offers detailed information on vehicle shape, positioning and orientation for vehicle area. It is distinct from conventional positioning systems which vehicles positioning based on single tag.
2) We design algorithms involving vehicle area calibration under both stationary and mobile conditions, including the extended Kalman filtering (EKF) algorithm and dynamic threshold algorithm, that fully utilize area positioning information. Experimental results show significant enhancements in positioning accuracy and stability. Additionally, both success warning rate and false warning rate have significant improvements compared to the positioning scheme employing the original two-way ranging (TWR) algorithm or single tag scenario.

In the remainder of this article, we present the overview of the proposed system in Section II. Section III details the positioning modeling method and the trajectory and steering calibration algorithm. In order to verify the effectiveness of our system and the superiority of our algorithm, we conduct experiments and data analysis in Section IV. Finally, we provide a summary of this work and suggest future directions in Section V.

## II. System Overview

In this section, we initially present the software, hardware structure, and deployment configuration of the proposed system. Subsequently, we provide a detailed explanation of the system's positioning and warning procedures. In contrast to conventional UWB positioning systems, our system excels in accurately modeling the location of vehicles within the designated area. Moreover, it facilitates real-time monitoring of the positioning results on the upper computer, offering an enhanced capability compared to traditional UWB systems.

### A. System Structure

As illustrated in Fig. 1, the system consists of several crucial components: positioning anchors, positioning tags, an upper computer, an warning module, and the target vehicle. The

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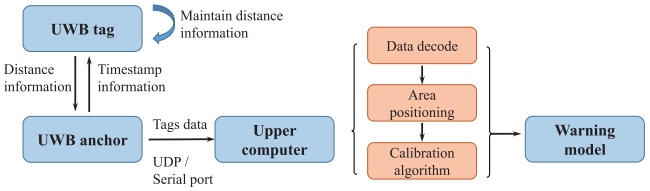



Fig. 2. Workflow of the proposed system.

architectural design of this system is inspired by the network system architecture delineated in [20].

Positioning anchors are strategically deployed along the periphery of the designated area, with a minimum of three or more anchors installed to maximize coverage. These anchors are categorized into two types: 1) slave anchors and 2) main anchor. Slave anchors are dedicated to communication with both the main anchor and positioning tags, as well as recording timestamps. The main anchor not only performs the functions of the slave anchors but also transmits data to the upper computer through a serial port, facilitating communication with the application layer.

Positioning tags are assumed to be placed in pockets or attached to safety helmets for pedestrian positioning, while for vehicles, tags are affixed at the four corners. These tags establish communication links with the anchors, capturing essential timestamp data and maintaining addresses and other critical information required for precise positioning.

The upper computer receives data transmitted via the serial port of the main anchor. Subsequently, it engages in data analysis, executes positioning algorithms, conducts optimization procedures, and generates visualized coordinate outputs.

The warning module can be installed within the vehicle and pedestrian both auditory and visual cues, such as a buzzer or LED, to deliver warning. When the warning module receives warning data transmitted from the upper computer, it activates the warning signal in accordance with the provided instructions.

### B. Work Procedure

The workflow of the system is illustrated in Fig. 2.

In the initial step, a minimum of three anchors are deployed to establish a planar positioning system. Each tag device within the system sequentially communicates with the anchors. With the TWR algorithm, the tag device calculates the round-trip time difference for each anchor and directly transmits the data to the main anchor.

Following this, the callback function of the main anchor receives positioning data from the tags. After executing the necessary format conversion, the data is transmitted to the upper computer through a serial port connection. The serial port interface can be linked to Wi-Fi modules, such as ESP8266, enabling wireless communication with the upper computer via the UDP protocol.

Finally, the upper computer initializes a thread to receive data from the main anchor. It decodes the received data, performs positioning calculations, executes optimization processes, and visually presents coordinate information. Additionally, the upper computer communicates with the warning module to determine whether to activate the warning as necessary.

## III. System Model

The determination of the vehicle area's location is achieved through UWB tags' positioning. To minimize positioning data errors and maintain the synchronization of area positions, it becomes essential to smooth and calibrate the real-time collected data. In this section, we present the positioning modeling method, along with the trajectory and steering calibration algorithm.

### A. Modeling of Pedestrian and Vehicle Area Positions

The positioning of pedestrians and vehicle areas necessitates the deployment of three or more UWB anchors. Utilizing distance measurements from three or more anchors to the tag, the trilateration algorithm can be employed to determine the tag's position, enabling the construction of a vehicle area model. The modeling of pedestrian and vehicle positions encompasses the following steps:

*1) Tag Location:* Existing positioning algorithms can be broadly categorized as time-based, including TWR and TDoA, and angle-based, such as Angle of Arrival (AoA) and signal strength-based positioning [21], [22], [23], [24], [25], [26]. Our system utilizes the TWR algorithm for positioning as its positioning accuracy and implementation effect aligns with our application requirements.

For the TWR positioning algorithm, the principle is to obtain the time $t_{tof}$ when the UWB pulse signal reaches each anchor, and then obtain the distance between the tag and each anchor as

$$r = t_{tof} \cdot c \tag{1}$$

where $r$ is the distance between tag and anchor, $t_{tof}$ is the time of UWB signal flight between them, and $c$ is the speed of light [27]. The geometric principle is shown in Fig. 4. In practice, in a positioning system with three anchors, the combination of two or more anchors can determine the intersection point of a circle, and the center of the intersection point of the three combinations is the current positioning point coordinate. The implementation of the TWR algorithm requires each tag to communicate sequentially with each anchor to obtain a round-trip timestamp instead of directly broadcasting, in order to determine its distance to each anchor. After obtaining the distance information of each anchor in the current scene, this information is then transmitted to the main anchor.

Specifically, in the TWR algorithm, the communication process between anchor $RX_1$ and tag $TX_1$ is as follows: $TX_1$ sends data $A$ frame to $RX_1$, and the timestamps for sending and receiving $A$ frame are $t_{t1}$ and $t_{r1}$. After receiving the data frame, $RX_1$ immediately replies to $TX_1$ with data $B$ frame, and the timestamps for sending and receiving $B$ frame are $t_{t2}$ and $t_{r2}$. After receiving $B$ frame, $TX_1$ sends $C$ frame again to $RX_1$, with timestamps $t_{t3}$ and $t_{r3}$ for sending and receiving $C$ frame. Due to the asynchronous time of $RX_1$ and $TX_1$, we only need

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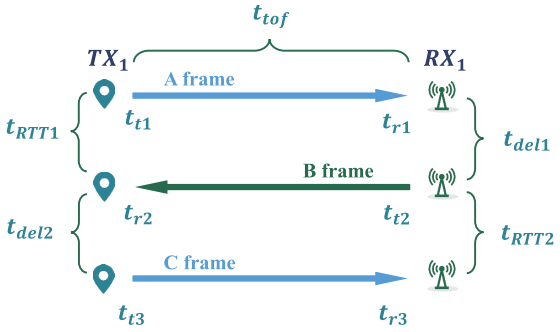



Fig. 3. Process of communication between tag and anchor in TWR.

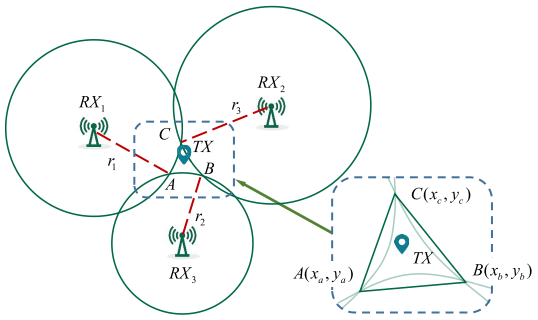



Fig. 4. TWR algorithm. The three points at the intersections of pairs of these circles form a triangle, and the centroid of the triangle is considered as the current coordinates for the tag.

to use their respective timestamps to calculate the round-trip time. As shown in Fig. 3, we can calculate the round-trip times of adjacent data as

$$\begin{aligned} t_{\mathrm{RTT1}} &= t_{r2} - t_{t1} \\ t_{\mathrm{RTT2}} &= t_{r3} - t_{t2}. \end{aligned} \tag{2}$$

Then, using timestamp information, we can calculate the transmission delay when two UWB devices send data as

$$\begin{aligned} t_{del1} &= t_{t2} - t_{r1} \\ t_{del2} &= t_{t3} - t_{r2}. \end{aligned} \tag{3}$$

We can obtain the time of flight between $TX_1$ and $RX_1$ as [28]

$$t_{tof} = \frac{t_{\mathrm{RTT1}} \cdot t_{\mathrm{RTT2}} - t_{del1} \cdot t_{del2}}{t_{\mathrm{RTT1}} + t_{\mathrm{RTT2}} + t_{del1} + t_{del2}}. \tag{4}$$

Now, we can obtain the distance between them based on (1), the distance between the remaining anchors and the tags can be determined using the same method. During this iteration of communication, $RX_1$ will send the distance information calculated from the previous iteration to $TX_1$ through frame $B$. We define the distances $r_1$, $r_2$, and $r_3$ between the tag and three anchors $RX_1$, $RX_2$, and $RX_3$, Due to measurement errors, circles with three anchors as centers and three measurement distances as radii may not exactly intersect at one point. We assume that the points that intersect pairwise and are closest to the center are $A(x_a, y_a)$, $B(x_b, y_b)$, and $C(x_c, y_c)$, their coordinates can be obtained as

$$\begin{aligned} (x_a - x_{RX1})^2 + (y_a - y_{RX1})^2 &= r_1^2 \\ (x_c - x_{RX1})^2 + (y_c - y_{RX1})^2 &= r_1^2 \\ (x_b - x_{RX2})^2 + (y_b - y_{RX2})^2 &= r_2^2 \\ (x_a - x_{RX2})^2 + (y_a - y_{RX2})^2 &= r_2^2 \\ (x_c - x_{RX3})^2 + (y_c - y_{RX3})^2 &= r_3^2 \\ (x_b - x_{RX3})^2 + (y_b - y_{RX3})^2 &= r_3^2. \end{aligned} \tag{5}$$

where $(x_{RX1}, y_{RX1})$, $(x_{RX2}, y_{RX2})$, and $(x_{RX3}, y_{RX3})$ are the coordinates of anchors $RX_1$, $RX_2$, and $RX_3$, which are constants in the scene. Therefore, by solving the six element equation system mentioned above, the position of $A$, $B$, and $C$ can be determined. To balance performance and accuracy, we opt for the centroid estimation algorithm to compute the coordinates' positions instead of the linearized least square estimation (LLSE) or fingerprint estimation (FPE) [29]. In order to make the tag positioning coordinate $(x_{TX}, y_{TX})$ more consistent with reality, we take the center of $\triangle ABC$ as its positioning coordinates, that is

$$(x_{TX}, y_{TX}) = \left( \frac{x_a + x_b + x_c}{3}, \frac{y_a + y_b + y_c}{3} \right). \tag{6}$$

*2) Real Time Modeling and Calibration of Vehicle Area:* With the acquired positioning data from each tag, the system advances to real-time modeling of the vehicle area. Utilizing a three-anchor setup allows for planar (2-D) positioning and modeling. The positioning data related to each tag is stored in the respective cache pool, facilitating area modeling and coordinate calibration. The cache pool adheres to an FIFO-like policy, meaning that as the capacity is reached, the earliest entered data will be flushed.

As shown in Fig. 5, the latest coordinate data of each tag are taken to form the original polygon, and the following operations are performed on the polygon.

1) Suppose the vertices of the polygon are $(x_i, y_i), i = 1, 2, \ldots, n$ according to (6), we define the center point of the polygon as $(x_{ctr}, y_{ctr})$, which can be calculated as

$$(x_{ctr}, y_{ctr}) = \left( \frac{\sum_{i=1}^{n} x_i}{n}, \frac{\sum_{i=1}^{n} y_i}{n} \right). \tag{7}$$

2) Calculate the offset angle of the polygon, where the offset angle of the first edge of the polygon is used as the overall offset angle of the polygon

$$\theta = \arctan \frac{y_2 - y_1}{x_2 - x_1}. \tag{8}$$

3) Generate standard polygon using the length and width information of the vehicle as

$$C_{std} = \begin{bmatrix} \left(-\frac{b}{2}, \frac{a}{2}\right), & \left(\frac{b}{2}, \frac{a}{2}\right) \\ \left(-\frac{b}{2}, -\frac{a}{2}\right), & \left(\frac{b}{2}, -\frac{a}{2}\right) \end{bmatrix} \tag{9}$$

where $a$ and $b$ are the length and width of the vehicle, respectively. $C_{std}$ is the corresponding standard polygon matrix, with each element representing its endpoint coordinates.

4) Rotate the standard polygon back to its actual angle, and add the offset value of the center point to obtain the calibrated polygon. We set the coordinate of each

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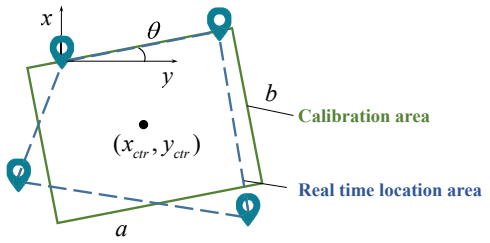



Fig. 5. Vehicle area modeling calibration.

vertex of the matrix as $(x_v, y_v)$, the coordinate matrix after rotation can be obtained as

$$\left[x_v^*, y_v^*\right] = \left[x_v - x_{ctr}, y_v - y_{ctr}\right]\begin{bmatrix}\cos\theta, -\sin\theta \\ \sin\theta, \cos\theta\end{bmatrix} + \left[x_{ctr}, y_{ctr}\right]. \tag{10}$$

5) Extend the modified polygon outward to the given safe distance (set to 2m in the experiment) to obtain a safe area polygon. Using this polygon, we can perform early warning judgment for pedestrians and vehicles.

After the aforementioned calibration operations, we obtain the latest area coordinates for each group. In comparison to single tag positioning, the positioning of the vehicle area becomes more accurate and reliable with the inclusion of information, such as center points and offset angles. By specifying a safe distance, a safety polygon can be constructed based on the positioning area. Utilizing collision detection algorithms, such as GJK [30], we can determine whether the pedestrian's positioning point or another vehicle's positioning area collides with the safety polygon. This process aids in determining whether the warning module needs to provide warning prompts.

### B. Trajectory and Steering Calibration

Given the inherent accuracy limitations of positioning firmware and data delays induced by channel conditions, addressing potential fluctuations and delays in positioning becomes crucial. Trajectory and steering calibration mechanisms are employed to ensure that real-time modeling aligns as closely as possible with the actual position. The specific steps involved are as follows:

*1) Extended Kalman Filtering Smoothing Trajectory:* EKF is often used for error compensation and smoothing of positioning coordinates in positioning system [31], [32], [33]. We fully utilize area positioning information for prediction and calibrate the next coordinate through the EKF algorithm, including the coordinate and velocity of the tag itself, as well as the yaw angle and yaw rate provided by the vehicle area. Let $\mathbf{Z}_t$ denote the measurement vector for each tag with sample time $t$. Hence, a state vector $\mathbf{X}_t$ is described as follows:

$$\mathbf{Z}_t = \left[x_t, y_t, v_t, \theta_t, \omega_t\right]^T \tag{11}$$

$$\mathbf{X}_t = \left[\hat{x}_t, \hat{y}_t, \hat{v}_t, \hat{\theta}_t, \hat{\omega}_t\right]^T. \tag{12}$$

where $(x_t, y_t)$ is the coordinate of the current tag, $v_t$ is the velocity, $\theta_t$ is the yaw angle of the current vehicle provided based on area positioning information and calibration by the algorithm in Section III-B2, and $\omega_t$ is the yaw rate. $(\hat{x}_t, \hat{y}_t, \hat{v}_t, \hat{\theta}_t, \hat{\omega}_t)$ are the corresponding predicted values. Further, the state equation is expressed as follows:

$$\mathbf{X}_t = \begin{bmatrix} \hat{x}_{t-1} + \frac{\hat{v}_{t-1}}{\hat{\omega}_{t-1}}\sin\left(\hat{\omega}\Delta t + \hat{\theta}_{t-1}\right) - \frac{\hat{v}_{t-1}}{\hat{\omega}_{t-1}}\sin\left(\hat{\theta}_{t-1}\right) \\ \hat{y}_{t-1} - \frac{\hat{v}_{t-1}}{\hat{\omega}_{t-1}}\cos\left(\hat{\omega}\Delta t + \hat{\theta}_{t-1}\right) + \frac{\hat{v}_{t-1}}{\hat{\omega}_{t-1}}\cos\left(\hat{\theta}_{t-1}\right) \\ \hat{v}_{t-1} \\ \hat{\theta}_{t-1} + \hat{\omega}_{t-1}\Delta t \\ \hat{\omega}_{t-1} \end{bmatrix} \tag{13}$$

where $\Delta t$ is the sample time. Specifically, when $\omega_{t-1} = 0$, the transfer formula becomes as follows:

$$\mathbf{X}_t = \begin{bmatrix} \hat{x}_{t-1} + \hat{v}_{t-1}\cos\left(\hat{\theta}\right)\Delta t \\ \hat{y}_{t-1} + \hat{v}_{t-1}\sin\left(\hat{\theta}\right)\Delta t \\ \hat{v}_{t-1} \\ \hat{\theta}_{t-1} + \hat{\omega}_{t-1}\Delta t \\ \hat{\omega}_{t-1} \end{bmatrix}. \tag{14}$$

In the EKF algorithm, the nonlinear equations of (13) and (14) are linearized based on the first-order Taylor expansion. It is given by the Jacobian matrix $\mathbf{J}$. Let $\mathbf{P}_t$ denote the covariance matrix of $\mathbf{X}_t$ which satisfies

$$\mathbf{P}_t = \mathbf{J}\mathbf{P}_{t-1}\mathbf{J}^T + \mathbf{Q} \tag{15}$$

where $\mathbf{Q}$ is the process noise covariance matrix. Let $\mathbf{K}_t$ is the Kalman gain vector, $\mathbf{R}$ is the measurement noise covariance matrix and $\mathbf{H}$ is measurement matrix. Thus, the procedure for EKF is shown in Algorithm 1. When the difference between the observed and predicted values is greater than $D_{\text{KFT}}$, it is considered abnormal. It needs to be filtered and reassigned to the predicted coordinate.

**Algorithm 1** EKF Smoothing Trajectory Algorithm

**Require:** $\mathbf{Z}_t$;
Initial state $\mathbf{X}_0$ and covariance matrix $\mathbf{P}_0$;
**for** $t = 0{:}\infty$ **do**
  // EKF Prediction
  Predict the state ahead $\mathbf{X}_t$ with (13) or (14);
  Predict the error covariance ahead $\mathbf{P}_t$ with (15);

  // EKF measurement update
  $\mathbf{K}_t = \mathbf{P}_t\mathbf{H}^T(\mathbf{H}\mathbf{P}_t\mathbf{H}^T + \mathbf{R})^{-1}$;
  $\mathbf{X}_t = \mathbf{X}_t + \mathbf{K}_t(\mathbf{Z}_t - \mathbf{H}\mathbf{X}_t)$;
  $\mathbf{P}_t = \mathbf{P}_t(\mathbf{I} - \mathbf{K}_t\mathbf{H})$;
**end for**
**if** $||\mathbf{Z}_t - \mathbf{X}_t|| > D_{\text{KFT}}$ **then**
  Filter the current coordinate and reassign as $\mathbf{X}_t$;
**end if**

*2) Dynamic Threshold Algorithm:* To further diminish the error in vehicle positioning during driving and fully leverage area positioning data for modeling and calibration, we propose an algorithm to obtain the positioning coordinates from the last instance for calibrating the vehicle steering and positioning. This algorithm is referred to as the dynamic threshold algorithm. The steps to update the vehicle positioning area using

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**Algorithm 2** Dynamic Threshold Algorithm

**Require:** $S_{\text{total}}$, $(x^*, y^*)$, $(x, y)$, $\overrightarrow{T}^*_{\text{pre}}$;
Compute $\overrightarrow{T}$ with (16);
Initializing the threshold $D_{\text{thre}} = 1$ ;
**if** $d > D_{\text{thre}}$ **then**
  Update $\overrightarrow{T}^*$ with (18) (cf. P3);
  $D_{\text{thre}} = D_{\text{thre}} + \Delta D$ (cf. P4);
**else if** $S_{\text{total}} < 1.0$ **then**
  $\overrightarrow{T}^* = (x_2 - x_1, y_2 - y_1)$ (cf. P5);
**else**
  $\overrightarrow{T}^* = \overrightarrow{T}^*_{\text{pre}}$ (cf. P6);
  $D_{\text{thre}} = D_{\text{thre}} - \Delta D/2$ (cf. P7);
**end if**
Unitize $\overrightarrow{T}^*$ as $\overrightarrow{T}$;
Update vehicle area vertex coordinates $C_{cor}$ using $\overrightarrow{T}$;
**return** $C_{cor}$, $\overrightarrow{T}^*$.

the dynamic threshold method in Algorithm 2 are summarized as follows.

**P1)** Calculate the vector from the center point coordinate of the previous time $(x^*, y^*)$ to the center point coordinate of that time $(x, y)$, and unit it (denoted as $\overrightarrow{T}$)

$$\overrightarrow{T} = \left(\frac{x - x^*}{d}, \frac{y - y^*}{d}\right) \tag{16}$$

$$d = \sqrt{(x - x^*)^2 + (y - y^*)^2}. \tag{17}$$

**P2)** Set initial threshold $D_{\text{thre}} = 1$.

**P3)** If it meets $d > D_{\text{thre}}$, the dynamic partitioning method is used to update $\overrightarrow{T^*}$, and the effect is that the faster the positioning coordinates change, the faster the direction changes

$$\overrightarrow{T^*} = \big(\mu(x - x^*), \mu(y - y^*)\big) + \overrightarrow{T}^*_{\text{pre}}(1 - \mu)$$
$$\mu = \alpha + \beta k \tag{18}$$

where $\overrightarrow{T}^*_{\text{pre}}$ is $\overrightarrow{T^*}$ calculated from the previous time which set to zero vector during algorithm initialization, $\mu$ is a polynomial about $k$, and different optimization effects can be obtained based on the value of the coefficient $\alpha$ and $\beta$. Besides, $k$ satisfies

$$k = \begin{cases} K, & d - D_{\text{thre}} > K \\ K/2, & K/2 < d - D_{\text{thre}} \le K \\ 0, & d - D_{\text{thre}} \le K/2. \end{cases} \tag{19}$$

Their values and performance will be discussed in Section IV.

**P4)** Update the coordinates of the endpoint of the vehicle area using the newly obtained vector as the turning direction of the vehicle at that time. In order to prevent the vehicle from constantly changing direction while driving, the threshold should be increased a given value $\Delta D$ to end the coordinate calibration.

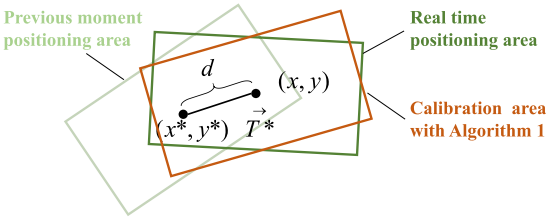



Fig. 6. Dynamic threshold algorithm ($d < D_{\text{thre}}$ & $S_{\text{total}} > 1.0$).

**P5)** Calculate the jitter variance of the positioning coordinates $S_{\text{total}}$ using the coordinate cache pool according to the following equation:

$$S_{\text{total}} = \sum_{i=1}^{n}\sum_{j=m}^{t-1}\left(\overrightarrow{z}_{ij} - \frac{\sum_{q=m}^{t-1}\overrightarrow{z}_{iq}}{t - m}\right)^2 \Big/ n \tag{20}$$

where $\overrightarrow{z}_{ij}$ represents the value of index $j$ in the current coordinate cache pool for the coordinates of the $i$th vehicle area polygon vertex, and $n$ is the number of vertices on the vehicle area polygon. If the coordinate jitter variance is less than a certain value (in this article is 1.0), it is considered that the vehicle has stopped at the time of positioning. At this time, there is no need to update the vehicle's steering based on the center point change vector $\overrightarrow{T}$, and there is no need to perform calibration in this iteration.

**P6)** In other situations, if the displacement of the vehicle is less than the threshold $D_{\text{thre}}$, we believe that the direction of the vehicle has not changed. At this point, the steering of the vehicle area is calibrated to last time's steering, as shown in Fig. 6.

**P7)** In addition, in order to prevent misjudgment caused by long-term slow movement of the vehicle, it is necessary to reduce the threshold for the next iteration of calibration judgment (in this article, the decrease of threshold is half the increase, i.e., $\Delta D/2$).

**P8)** After completing calibration, we add the final coordinates to the coordinate cache pool for reference in the next time of coordinate calibration.

By implementing the above calibrations, significant improvements have been achieved in accurately localizing the vehicle area under both stationary and dynamic conditions.

## IV. Numerical Analysis and Experimental Results

In this section, we quantitatively evaluate the effectiveness of the proposed system. All experiments are conducted on-site with data collected for comparative experiments. The parameters and their values are listed in Table I, the experimental scene is shown in Fig. 7, and the deployment of tags on vehicle and pedestrian is shown in Fig. 8.

### A. Trajectory Capturing

To visually showcase the effectiveness of the positioning system, we present the actual trajectory and positioning coordinates of the vehicle in both driving and stationary situations, as shown in Figs. 9 and 10.

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TABLE I
MEASURED PARAMETER VALUES OF THE AREA POSITIONING SYSTEM

| Parameter | Value |
|---|---|
| **Anchor coverage** | Scenario 1: 75m×38m<br>Scenario 2: 46m×15m |
| **Upper computer configuration** | i7-8750H 1050ti |
| **UWB antenna model** | DW1000 |
| **Single Tag positioning frequency** | 10Hz |
| **Number of UWB Tags** | 5 (4 vehicle tags<br>1 pedestrian tag) |
| **Number of UWB Anchors** | 3 |
| **Vehicle safety distance** | 2m |
| **Vehicle area size** | 1.5m×3.0m |

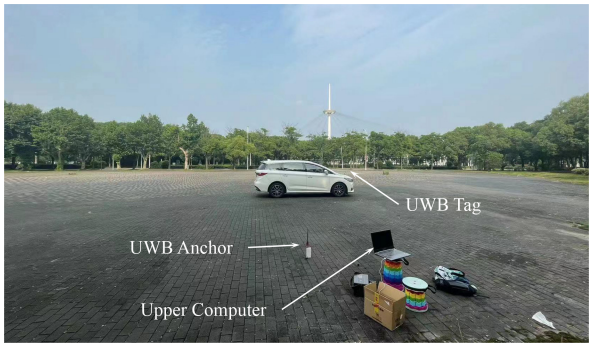



Fig. 7. Experimental site for positioning system (Scenario 1).



Fig. 8. Deployment of tags on vehicle and pedestrian.

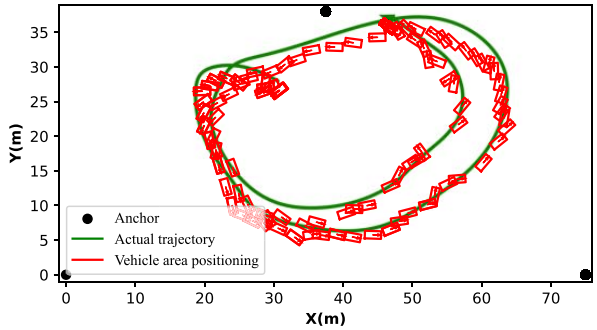



Fig. 9. Actual and positioning trajectories (the arrow on the rectangle indicates the direction of the front of the vehicle, the $x$ and $y$ coordinates represent the length and width of the experimental site, the same below).

For Scenario 1, the continuously gathered data of vehicle area positioning are illustrated as the rectangle in Fig. 9, along with the actual vehicle trajectory represented by a curve. The ground truth data is fitted from visual and sensor data. It is evident that the vehicle area's positioning aligns well with the actual vehicle route. The results indicate that the trajectory and steering calibration algorithm during driving conditions facilitates the consistency between the calibrated and actual orientation during driving conditions.

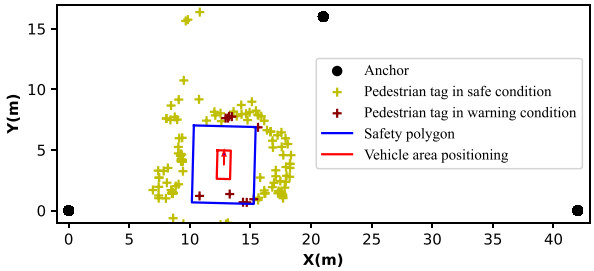



Fig. 10. Positioning of pedestrians walking around a safe distance when the vehicle is stationary (overlay at different times).

Fig. 10 illustrates a case where a pedestrian is moving around a stationary vehicle beyond the safety distance in Scenario 2. The outer rectangle represents the safety distance, and the dots correspond to the collected positioning of pedestrian tags at different times. The pedestrian tag moves randomly around the vehicle. If the pedestrian approaches within an unsafe distance, an alert will be triggered, warning the pedestrian to move away. Due to the implementation of the trajectory and steering calibration, the steering of the vehicle positioning area remains stable even when the vehicle is stationary, minimizing fluctuations in the tag coordinates. As a result, the majority of pedestrian coordinates are located outside the designated polygon, indicating that the implementation of trajectory and steering calibration algorithms helps reduce the false warning rate.

In addition, we conducted warning performance tests on the proposed system in dynamic scenarios, as shown in Fig. 11. Pedestrian and vehicle are located at positions (7m, 0) and (7m, 10m) on the left side of the scenario, moving clockwise and counterclockwise along the rectangular trajectory. It can be seen that when the pedestrian enters the warning area of the vehicle, the system will determine that the pedestrian's coordinates are in the warning condition, and then provide a warning prompt. In addition, we tested the success rate and false alarm rate of warnings in dynamic scenarios, where pedestrians were allowed to move with the vehicle within and outside the warning area, with probabilities of 95.21% and 6.33%, respectively. It can be seen that the proposed system also has good warning performance in dynamic scenarios.

### B. Positioning Accuracy Analysis

To evaluate the positioning accuracy of our system, we measured the root mean square error (RMSE), STD and the cumulative distribution function (CDF) of the proposed system, and provided measurement data for single tag and vehicle area (multitag), the results are shown as in Table II and Fig. 14. The positioning trajectories are shown in Figs. 12 and 13, respectively. In order to facilitate data analysis, in the experiment, we moved the single tag and vehicle area counterclockwise along the black rectangular trajectory at a speed of 5km/h in the figures, and analyzed the differences between the positioning results and the actual trajectory.

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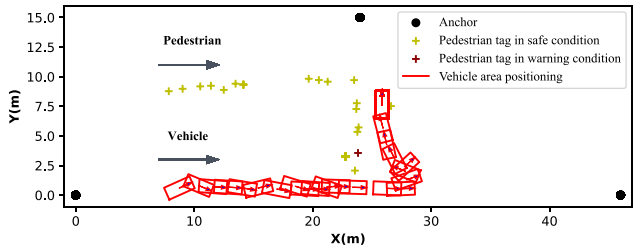



Fig. 11. Dynamic scenario warning situation (pedestrian and vehicle moving clockwise and counterclockwise, respectively, from the left, triggering warning prompts when they intersect).

TABLE II
SYSTEM POSITIONING ACCURACY MEASUREMENT DATA

| Measurement target | RMSE | STD |
|---|---|---|
| Single tag | 0.5655 | 0.4619 |
| Calibrated single tag | **0.4644** | **0.4152** |
| Vehicle tag1 | 0.7369 | 0.9000 |
| Vehicle tag2 | 0.6691 | 0.8719 |
| Vehicle tag3 | 1.0822 | 1.2856 |
| Vehicle tag4 | 0.4994 | 0.7720 |
| Calibrated vehicle area center | **0.4467** | **0.3496** |

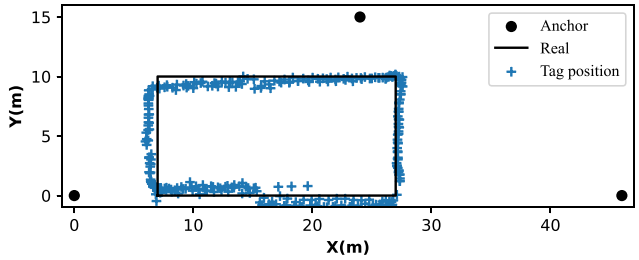
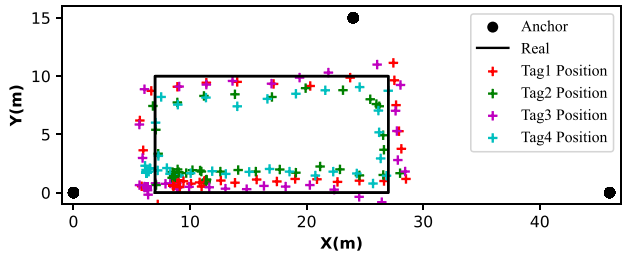



Fig. 12. Positioning scenario of single tag moving along trajectory.

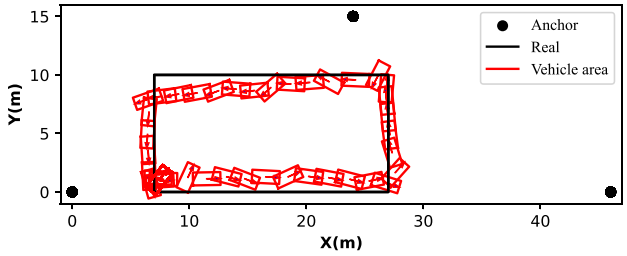
From Table II, it can be seen that the RMSE and STD of the calibrated single tag coordinates are 17.89% and 10.10% smaller than the original data, respectively. In addition, due to the presence of redundant information from multitag, the center point's RMSE and STD of area positioning are also smaller than those of single tag positioning. Compared to each vehicle tag, the RMSE of area positioning decrease by 67.22% and the STD decrease by 63.47%. From the CDF data in Fig. 14, it can be seen that most positioning point errors are within 0.6m, and the CDF of the area center is steeper near 0m than single tag, indicating that there are more error values distributed around 0m. In summary, the coordinates calibrated by the optimization algorithm will be more in line with the real trajectory compared to the original coordinates.

### C. Performance of Optimization Algorithms

To evaluate the performance of the proposed optimization algorithm, we have conducted multiple tests with different parameters. For the proposed system, we focus on the stability of positioning, which is a key improvement in algorithm optimization. Therefore, we evaluate its performance using the magnitude of vehicle body shake variance and RMSE. The




(a)




(b)

Fig. 13. Vehicle area positioning situation: (a) Vehicle tags and (b) vehicle area after calibration algorithm.

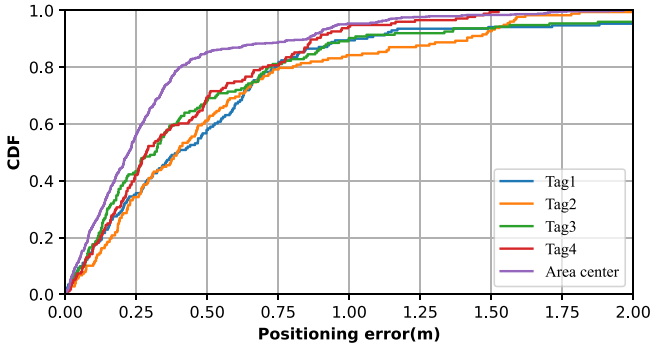



Fig. 14. Positioning error CDF of vehicle tags.

definition of vehicle body shake variance $V_{vbs}$ is as

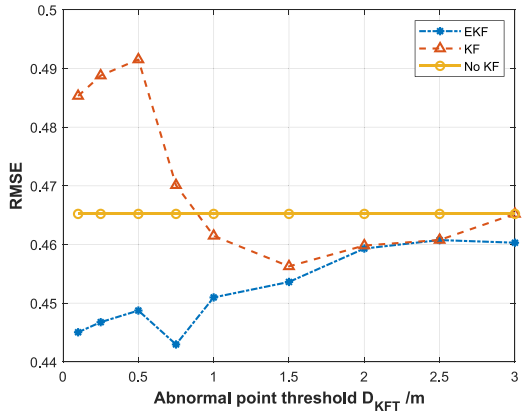
$$V_{vbs} = \sum_{i=m}^{t-1} \left( \arctan \overrightarrow{T}_i - \frac{\sum_{j=m}^{t-1} \arctan \overrightarrow{T}_j}{t-m} \right)^2 \tag{21}$$

where $\overrightarrow{T}_i$ represents the value of index $i$ in the current coordinate cache pool for the unit vehicle steering vector. The results are shown in Figs. 15 and 16.

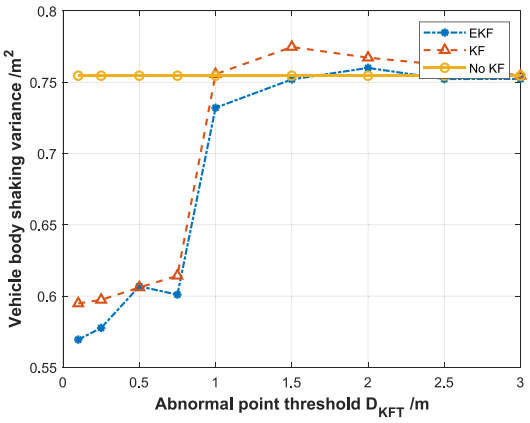
*1) Stability With Different $D_{\mathrm{KFT}}$:* In order to test the impact of different $D_{\mathrm{KFT}}$ values on the Kalman filtering performance of the proposed positioning system, as well as the impact of the proposed EKF on positioning accuracy and stability, we designed comparative experiments in different situations. The experimental scenario is shown in Fig. 13. We processed the data using EKF, ordinary Kalman filtering (i.e., using only coordinate data for linear prediction), and no Kalman filtering to measure the vehicle body shake variance and RMSE of the positioning system. When discussing the performance of EKF, the dynamic threshold algorithm will not be enabled.

The results are shown in Fig. 15. This is because the smaller the $D_{\mathrm{KFT}}$ value, the more coordinate values filtered

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(a)




(b)

Fig. 15. Indicator under with variation of $D_{\mathrm{KFT}}$ in different case: (a) RMSE and (b) vehicle body shake variance (case 1: EKF, case 2: KF, and case 3: not using filtering).

and reassigned by the Kalman filter in Section III-B1, and the greater the difference between its values and the original data, resulting in greater fluctuations in vehicle body turning. On the other hand, the larger the $D_{\mathrm{KFT}}$ value, the closer the coordinate value is to the original data. The vehicle body jitter and RMSE under two Kalman filters both approach the original data as $D_{\mathrm{KFT}}$ increases. From the RMSE data, it can be seen that in terms of positioning accuracy, the EKF of this system, due to the use of redundant information from multitag, has higher accuracy than the ordinary KF at different thresholds, and the vehicle body jitter is also smaller than that of the ordinary KF. Choosing an appropriate $D_{\mathrm{KFT}}$ value is crucial, as an excessively small $D_{\mathrm{KFT}}$ may misclassify some normal positioning points during vehicle movement as abnormal, causing a deviation from the true position. Conversely, an excessively large $D_{\mathrm{KFT}}$ may result in the Kalman filter failing to successfully reassign some abnormal points, emphasizing the importance of selecting an optimal $D_{\mathrm{KFT}}$ value in practical applications.

*2) Stability With Different Parameter Values in Dynamic Threshold Algorithm:* For the dynamic threshold algorithm proposed in Section III-B2, the values of parameters $\alpha$, $\beta$, and $K$, as well as the magnitude of $D_{\mathrm{thre}}$ changes $\Delta D$ in each iteration, affect the calibration of the vehicle's steering. Therefore, we conduct multiple tests to discuss the overall performance of the dynamic threshold algorithm by fixing other parameters and changing a certain value.

The performance of the algorithm with different parameter values is illustrated in Fig. 16. The impact of $\Delta D$ in each iteration on the vehicle's steering oscillation is observed to be limited. However, the selection of different values significantly influences scenarios involving extreme steering oscillation. A larger magnitude of $D_{\mathrm{thre}}$ change leads to quicker transitions of the algorithm between P3 and P5. Consequently, during substantial turning movements of the vehicle, the oscillation becomes more pronounced. Therefore, in practical applications, it is advisable not to choose a high-variation magnitude to prevent excessive steering responses to significant vehicle maneuvers, such as turns.

Additionally, it can be observed that the increase of $\alpha$ and $\beta$ results in more pronounced steering oscillation of the vehicle. This is attributed to the combined influence of these parameters on the steering during vehicle movement. With other parameters fixed, larger values of $\alpha$ and $\beta$, as indicated by (18), lead to steering calibration that approaches the displacement $\overrightarrow{T}$ from the previous to the current time step on the central coordinate point. Due to inaccuracies in the original positioning data, there might be some deviation in $\overrightarrow{T}$ during vehicle movement. Consequently, the steering oscillation becomes more noticeable. In practice, it is advisable to avoid excessively high values for $\alpha$ and $\beta$ to mitigate the prominence of steering oscillation.

Finally, the significance of the parameter $K$ lies in steering oscillation converging toward statistical stability. With an increase in $K$, the steering oscillation exhibits a trend of initially increasing and then decreasing. For instance, when comparing $K = 0.1$ and $K = 30$, the vehicle body shake variance remains relatively consistent ($0.045\mathrm{m}^2$ and $0.037\mathrm{m}^2$). This behavior is attributed to the fact that, during vehicle movement, i.e., when the algorithm is in P3, it can be inferred from (19) that, whether $K$ is large or small, the contribution of the displacement vector at this moment to the update of the vehicle's heading vector in (18) is relatively small. As a result, the steering remains more stable during movement.

### D. Impact of Calibration

In this section, we compare the indicator differences of the positioning system with traditional single tag scenario (vehicle positioning based on single tag), original area scenario and calibrated area scenario. The single tag scenario refers to deploying the tag in the center of the vehicle, and determining whether a warning prompt is issued by determining whether the distance between the pedestrian tag and the vehicle tag is less than a safety distance (in this article, it is set as the sum of half the circumference of the car polygon and the vehicle safety distance). The calibration here includes EKF and dynamic threshold algorithm calibration. We select a set of well-performing calibration algorithm parameters to participate in the experiment, with specific values as shown

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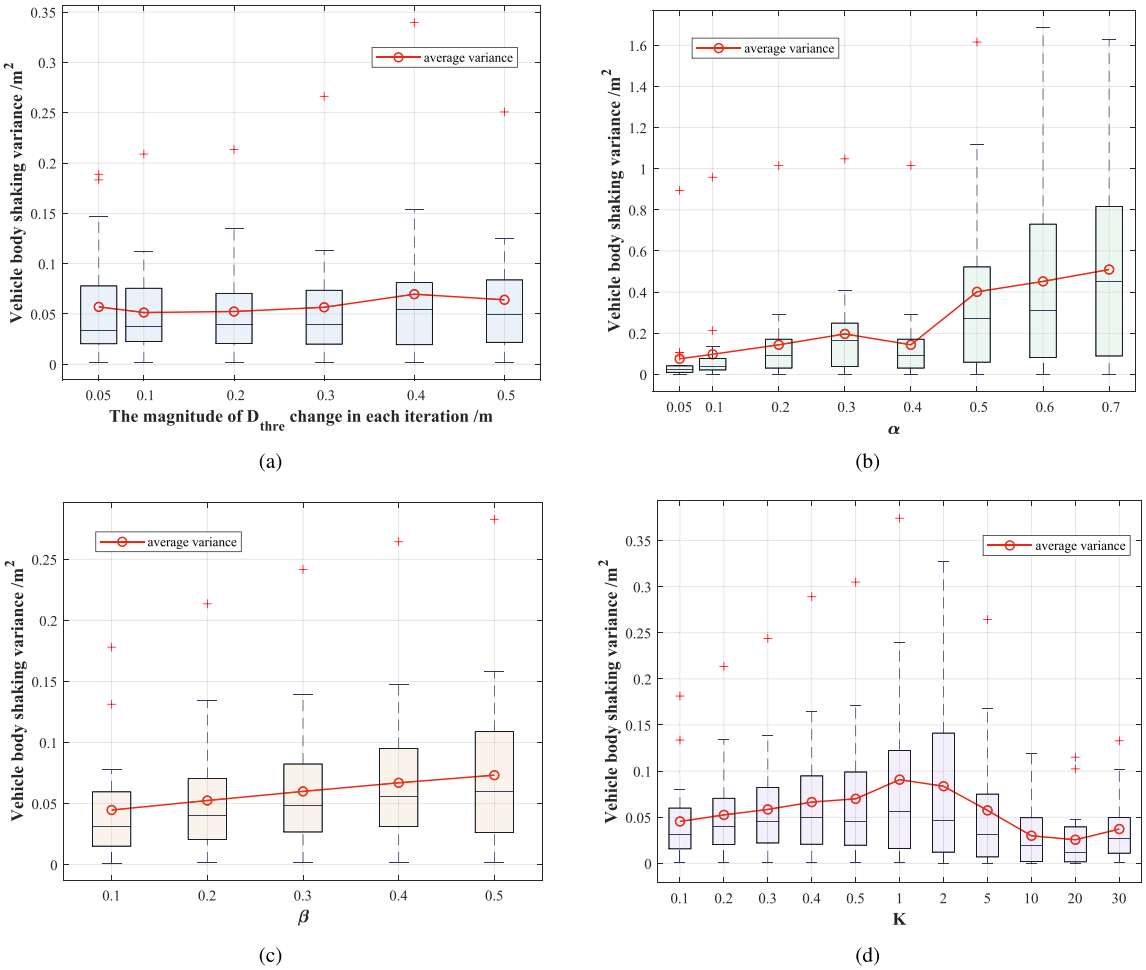



Fig. 16. Performance of vehicle body shake variance under different parameter values in the dynamic threshold algorithm: (a) Different magnitude of $D_{\text{thre}}$ ($\Delta D$) change ($\alpha = 0.1, \beta = 0.2, K = 0.2$); (b) different $\alpha$ ($\Delta D = 0.1, \beta = 0.2, K = 0.2$); (c) different $\beta$ ($\Delta D = 0.1, \alpha = 0.1, K = 0.2$); and (d) different $K$ ($\Delta D = 0.1, \alpha = 0.1, \beta = 0.2$).

TABLE III
CALIBRATION ALGORITHM PARAMETERS

| Parameter | Symbol | Value |
|---|---|---|
| **Kalman filtering threshold** | $D_{KFT}$ | 0.5m |
| **The magnitude of $D_{thre}$ change** | $\Delta D$ | 0.1 |
| **Steering parameters** | $(\alpha, \beta, K)$ | (0.1, 0.2, 0.2) |

in Table III, and the comparison results are shown in Figs. 17 and 18.

In Fig. 17, we overlay the original and calibrated positioning area at different time. It can be observed that the calibrated vehicle positioning area turns smoother and does not experience significant fluctuations due to errors in the original positioning data.

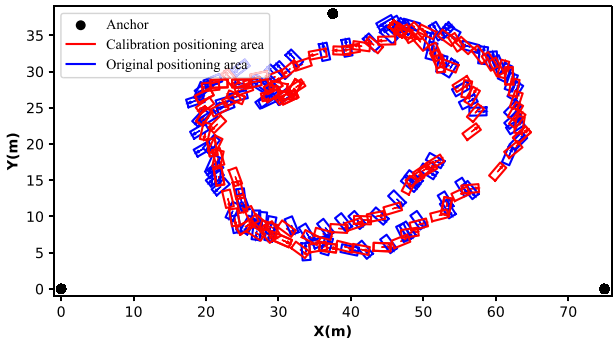



Fig. 17. Comparison of overlay diagram of vehicle body steering at different times for before and after calibration.

To illustrate the impact of calibration algorithms on the stability of positioning system, we compared the indicators of STD and vehicle body shake variance. The results presented in Fig. 18 show that after trajectory and steering calibration, the body shake caused by positioning point errors during driving has been reduced to 81.57%. Additionally, when the vehicle is stationary, the STD of coordinate decreases to 49.44% of the original data. It can be concluded that, the positioning area is more stable and the steering is smoother compared to the original data.

Regarding positioning errors, we separately calculated the success rate and false warning rate for single tag data, the original area data and the calibrated data. Due to the lack of vehicle shape and angle information in single tag positioning, the warning effect is worse than in multitag vehicle area positioning. In addition, due to the reduction in positioning

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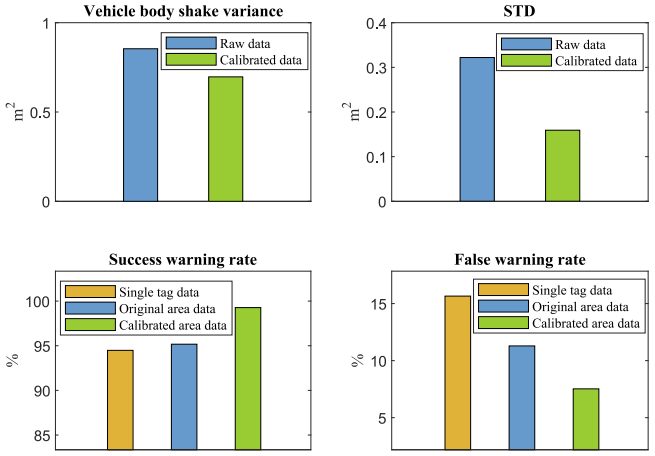



Fig. 18. Comparison of indicators for traditional single tag scenario, original area scenario, and calibrated area scenario.

coordinate STD caused by calibration and the improved stability of the vehicle body, the warning success rate increased by 4.30% when the pedestrian tag enters the safety polygon in the area. Similarly, when the pedestrian tag is outside the safety polygon of the area, the false warning rate is reduced by 33.35%.

## V. Conclusion

In this article, we introduce a UWB-based positioning system designed for precise pedestrian and vehicle area positioning and warning. Therefore, the system shows potential to improve safety in complex scenarios involving interactions between pedestrians and vehicles.

Our proposed system incorporates improvements by adjusting area positioning coordinates based on prior data. We employ trajectory and steering calibration algorithms, including EKF and the dynamic threshold algorithm, to compensate for positioning errors, bringing the positioning area into closer alignment with real-world conditions.

The results demonstrate notable enhancements, with the proposed system reducing positioning STD by 18.43% and vehicle body shake variance by 50.56%. Additionally, the warning success rate increased by 4.30% and the false warning rate is reduced by 33.35%. Meanwhile, we have explored and discussed the performance of optimization algorithms under varying parameter values.

For future work, we aim to further enhance the system's robustness, such as incorporating an inertial measurement unit (IMU) fusion positioning method, to address positioning errors and NLoS scenarios. Additionally, we plan to explore the application of machine learning techniques to differentiate between LoS and NLoS signals.

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